object.size()

The *object.size()* function gives an estimate for the memory used by an object. It accepts an R object as its argument. You can also pass it a logical value for 'quote' to indicate if the result should be displayed within quotes. The 'units' argument can specify the units for printing the values. The following standards for the units are accepted – 'legacy', 'IEC', and 'SI'. A numeric data type consumes 48 bytes, and with the data there is an additional 8 bytes of overhead.

```
> object.size(numeric())
48 bytes
> object.size(2)
56 bytes
> object.size(m)
3600216 bytes
```

Rprof and summaryRprof

The time spent by various R functions can be displayed using the *Rprof()* function. It takes the following arguments:

Argument	Description
filename	Output filename to store results. Use NULL to stop profiling
append	Logical value to append results or overwrite the file
interval	The time interval between taking samples. Default is 0.02
memory.profiling	Logical value to write memory information to output file

You can create a 'profile.out' results file using the *Rprof()* function to profile the *crossprod()** function, as illustrated below:

```
> Rprof(filename = "profile.out", append = FALSE, interval =
0.03, memory.profiling=TRUE)
> crossprod(m, n)
   [,1]       [,2]          [,3]          [,4]          [,5]
    [1,]  1.596159e+02  -8.97348864 -7.103962e+01
3.863508e+01 -1.168697e+02
    [2,] -3.959816e+01   13.93104995 -7.210794e+01
3.495264e+01  4.697674e+01
    [3,]  1.329374e+02  -80.40730081  5.745854e+01
-7.841128e+01 -8.069322e+01
    [4,]  7.727353e+00   78.42727385  1.142679e+01
-9.473472e+01 -4.733995e+01
    [5,]  1.341396e+02  -50.23056812  7.478321e+01
-5.869277e+01 -9.588495e+00
    [6,]  1.741993e+01   45.03771290  5.569390e+01
-2.684412e+01  5.295564e+01
...
> Rprof(NULL)
```

After the profiling is stopped by using the 'Rprof(NULL)' command, the results can be viewed using the *summaryRprof()* function:

```
> summaryRprof(filename = "profile.out")
$by.self
            self.time self.pct total.time total.pct
"crossprod"      3.96      100       3.96       100

$by.total
            total.time total.pct self.time self.pct
"crossprod"       3.96       100      3.96      100

$sample.interval
[1] 0.03

$sampling.time
[1] 3.96
```

Since we are only running the 'crossprod()' function, the output shows that it takes all the time in the computation. The 'summaryRprof()' function accepts the following arguments:

Argument	Description
filename	The file generated by *Rprof()*
memory	Memory information
lines	Line information
index	Stack trace memory information
diff	Logical value to indicate the difference in memory
exclude	Any functions to filter out in the output

The *memory="stat"* option to the *summaryRprof()* function displays the maximum memory usage, as shown below:

```
> summaryRprof(filename = "profile.out", memory="stat")
index: "crossprod"
     vsize.small  max.vsize.small      vsize.large  max.
vsize.large
           11749          1597864         13309000
1799964112
```